JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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NAAC "A" Grade Department of Basic Science & Humanities

Visio n	Missio n
To lay a robust foundation for the Institute to reach its Zenith.	▪ Achieving academic excellence through rigorous teaching,learning and evaluation practices. ▪ To develop an ability to apply knowledge of basic scienceand Mathematics to excel in the field of Engineering. ▪ To provide salutary environment for the betterment offaculty and the students.

Branch : ETC / ITB

Subject : Advanced Chemistry

Unit-4 A) Atomic and molecular structure

Dalton's Atomic Theory :-

1. Elements consist of indivisible small particles called atoms.
2. All atoms of same elements are identical; different elements have different types of atoms.
3. Atoms can neither be created nor destroyed.
4. Atoms of different elements combine in fixed whole number ratios when forming compounds.

Rutherford Atomic Model :-

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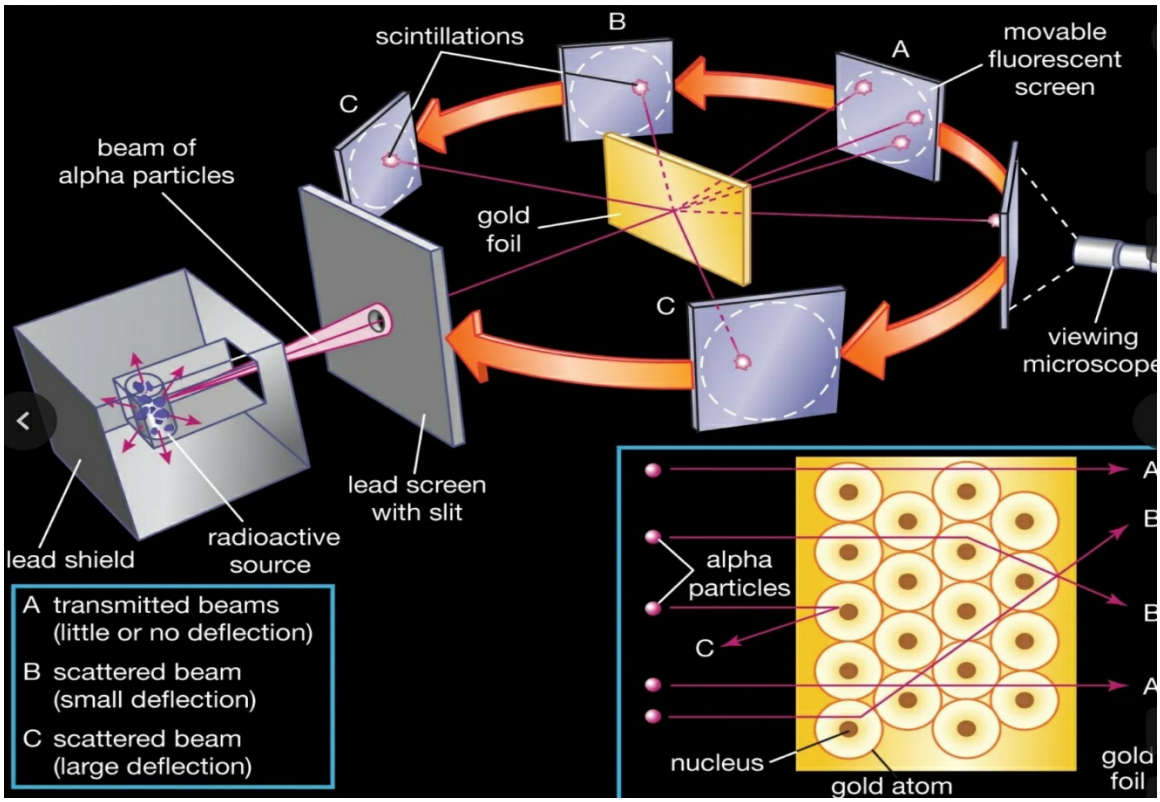

1. Atom has a tiny positively charged central part known as ‘nucleus’.
2. Most of the volume of atoms is empty.
3. Almost the whole mass of an atom are concentrated in the nucleus.
4. Electrons revolve around the nucleus in the extra nuclear space in various orbits, which are called as shells or energy levels.
5. Number of electrons is equal to the number of protons.
6. Except electrons, all the fundamentals particles are present in the nucleus and are called nucleons.

Drawbacks : It does not explain how the atom is stable. As per Rutherford, electrons are revolving around the nucleus is an accelerating charge particle. According to the electromagnetic theory, an accelerating charged particle emit radiations, and lose energy. Because of the loss of energy, the electron would slow down, and will not be able to withstand the attraction of the nucleus. As a result, the electron should follow a spiral path, and ultimately fall into the nucleus. If it happens then the atom should collapse. But this does not happen: atoms are stable.

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Bohr's Atomic model :-

1. In an atom electrons are present as tiny particle.
2. Electrons are revolve around the nucleus in fixed circular path called orbit/ energy shell/energy level.
3. Electron revolve in an orbit maintain fixed distance from the nucleus.
4. Energy level represented by as K, L, M, N called principal quantum number i.e., 1, 2, 3, 4 and so on.
5. According to Bohr, if electron revolving in particular orbit if it maintain its path, it neither absorb nor emits energy.
6. Hence, these energy levels are called stationary states energy

$$\Delta E = E2 - E1 = h\vartheta$$

Angular momentum of electron $mrv = n\frac{h}{2\pi}$

Drawbacks :

1. could not explain spectral behavior of multi electron species.
2. Not able to explain splitting of spectral line in presence of magnetic field and electric field.

Quantum mechanical model :

de Broglie : Electrons have wave like property.

heseinberg uncertainty principle : we can't describe the exact position and momentum of an electron.

schrodinger's wave equation :

$$H\varphi = E\varphi$$

It treated electron as a wave in a atomic model and it is called as quantum mechanical model.

Bohr's orbit are replaced by quantum mechanical orbital.

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Concept of orbital:

Orbital : orbital represents probability map that shows distribution of electron to be found around the nucleus.

Lowest energy orbital – 1s

Quantum number

1. **Principal quantum number (n) :**

   It indicate most probable position where electron revolve / found around the nucleus.

   n = 1, 2, 3, 4, …

   n = 1 K shell

   n = 2 L shell

   n = 3 M shell

   n = 4 N shell

2. **Azimuthal quantum number (l) :**

   It indicate subshell / sublevel l = (n – 1)

   If n=1, then l = 1-1 = 0 s subshell

   If n=2, then l = 2-1 = 1 p subshell

   If n=3, then l = 3-1 = 2 d subshell

   If n=4, then l = 4-1 = 3 f subshell

3. **Magnetic quantum number (m) :**

   Indicates magnetic properties of an electrons

   m = 2l + 1

   If l = 0 then,

   m = 2 ×0 + 1

   m = 1

Atomic orbitals : s, p, d, f

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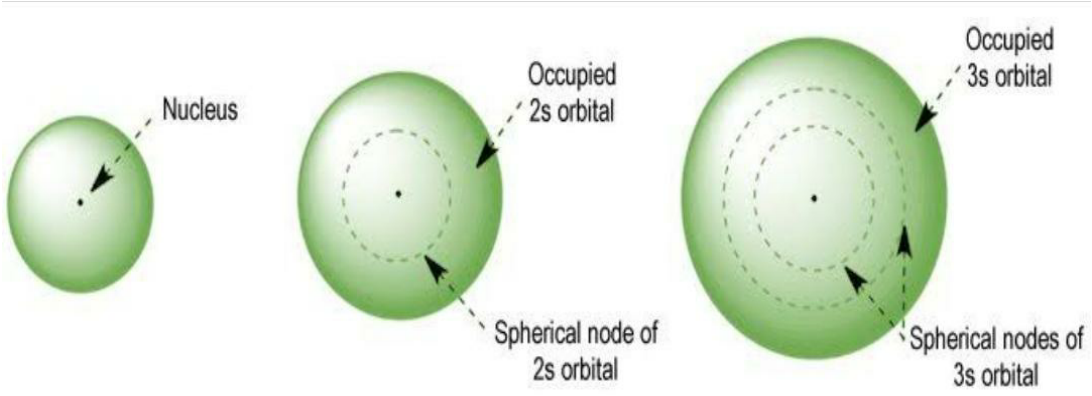
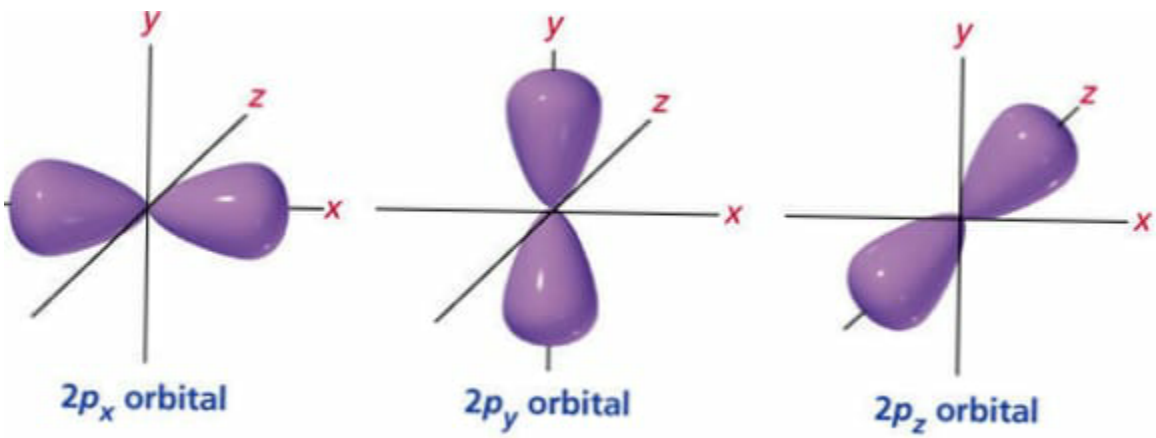
Shape of orbitals :

s- orbital : spherical shape



p- orbital : dumbbell shape : p_x, p_y, p_z



d - orbital : double dumbbell shape : dxy, dyz, dxz, dx^2-y^2, dz^2

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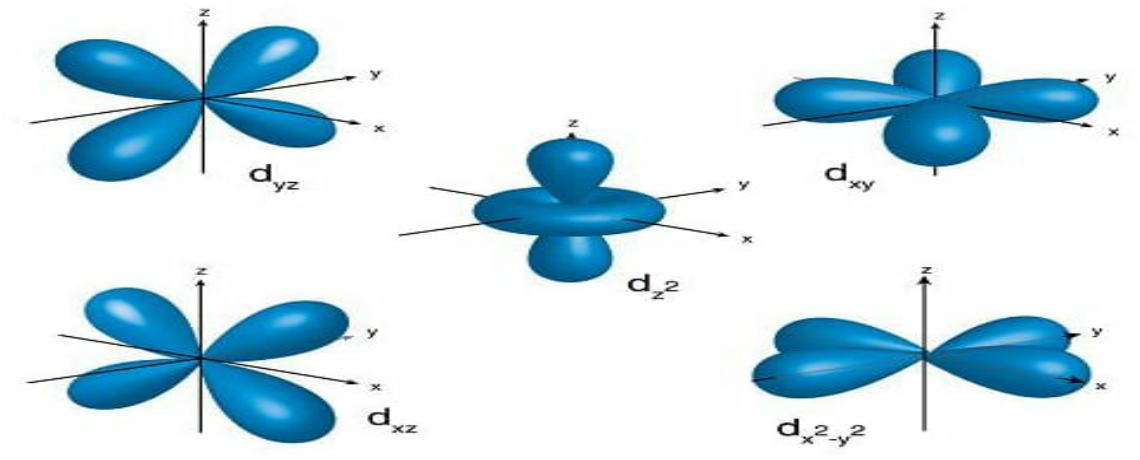

4. **Spin quantum number (s)** : It indicates spin of an electrons

   s = +1/2 , -1/2

Pauli's exclusion principle :

" No two electrons in an atom can have the same values of all the four quantum numbers."

According to this principle no two electrons in same atom can have same values for the four quantum number, i.e., n, l, m, s. It means that if two electrons have identical values of n, l, and m they must differ in spin quantum numbers. According to this principle the spin quantum number values should be different i.e., s = +1/2 and s = -1/2. Any orbital can accommodate a maximum of two electrons with opposite direction ( )
.

Hunds rule of maximum multiplicity :

" When the electrons are entering into different orbital from the same energy level, they enter singly with parallel direction of spin before pairing starts."

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Difference between orbit and orbital

Orbit	Orbital
An orbit is a fixed path on which electrons revolve around the nucleus.	An orbital is the probable area of finding the maximum density of electrons in an atom.
An orbit is a planar representation, i.e., a two dimensional representation.	An orbital is a three dimensional representation.
An orbit is non-directional in nature which means the shape of an atom cannot be described by an orbit.	While an orbital can describe the shape of an atom thus is directional in nature.
An orbit does not follow the theory of Heisenberg's Uncertainty Principle. Orbits follow the principles of Bohr-Sommerfeld's theory.	An orbital follows the theory of Heisenberg's Uncertainty Principle.
An orbit can accommodate electrons where n represents the number of the orbit or the shell. For example, K shell represents the 1st orbit, L shell represents the 2nd one.	An orbital can accommodate the maximum of two electrons only in its sub-levels. The s orbital has only one sub-level, so it can contain only 2 electrons. But the p orbital has 3 sub-levels and thus it can contain upto 6 electrons.
Orbits are circular or elliptical.	Orbitals have different geometrical shapes i.e., spherical, dumb-bell shaped.

MOLECULAR ORBITAL THEORY

- The atomic orbitals overlap to form new orbitals called molecular orbitals. When two atomic orbitals overlap, they lose their identity and form new orbitals called molecular orbitals.
- The electrons in the molecules are filled in the new energy states called the molecular orbitals, similar to the electrons in an atom being filled in an energy state called atomic orbitals.
- The probability of finding the electronic distribution in a molecule around its group of nuclei is given by the molecular orbital.
- The two combining atomic orbitals should possess energies of comparable value and similar orientation. For example, 1s can

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combine with 1s and not with 2s.

- The number of molecular orbitals formed is equal to the number of atomic orbitals combined.
- The shape of molecular orbitals formed depends upon the shape of the combining atomic orbitals.
- According to the molecular orbital theory, the filling of orbitals takes place according to the following rules:

Aufbau's principle: Molecular orbitals are filled in the increasing order of energy levels.

Pauli's exclusion principle: In an atom or a molecule, no two electrons can have the same set of four quantum numbers.

Hund's rule of maximum multiplicity: Pairing of electrons doesn't take place until all the atomic or molecular orbitals are singly occupied.

Linear combination of atomic orbitals approximation formation of bonding and antibonding molecular orbitals.

The wave function for molecular orbitals is obtained by a method known as linear combination of atomic orbitals (LCAO) approximation. In this method the linear combination of two atomic orbitals wave function is brought about either by adding orbits obstructing the wave functions thus if c is the wave function of the molecular orbital obtained from two atomic orbitals of two atoms A and B having ψ_A and ψ_B respectively as their function.
Then

Molecular Orbitals of H_2

The hydrogen atom is the simplest atom, and its molecule H_2 is also the simplest molecule than monoatomic molecules such as He and Ne, etc. Because of their simplicity, they have been extensively studied. Many theories have been developed using them as models.

The Hydrogen Molecule

Atomic hydrogen gas is unstable, existing only at very high temperature. Diatomic hydrogen molecules, H_2 are formed at ordinary temperatures and pressures. Many models and theories have been proposed to explain the chemical bond, and here are some simplified forms of these theories, some of which may have been proposed as society began to speculate what is happening. A brief review is given

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below, and some of these items will be further explained.

- The hydrogen atom has a valency of 1 and it tends to form a chemical bond with another atom. It forms a covalent bond with another atom of the same kind to form H−H
- When people realized that the hydrogen atom has one electron, and that He has two electrons, they thought that an atom has a tendency to share electrons to achieve a closed shell as those of inert gases. Thus, its structure should be represented by H:H the Lewis dot structure.
- The invention of quantum mechanics to describe the electron configuration of atoms led to the electron configuration of $1s^1$ for hydrogen. This electron tends to find a partner, and sharing an electron with another hydrogen atom seems to make it more stable.
- Atoms overlap to form a chemical bond.
- Molecular orbital (MO) theory suggested atomic orbitals which not only overlap, but are simultaneously transformed into molecular orbitals with new energies and new electron distributions.

Some facts are known about hydrogen due to some careful experimental measurements. The internuclear distance is 74 pm, and the dissociation of an H−Hbond H bond into two atomic hydrogen atoms requires 7.2×10^{-19} J, (usually given as 435.9 kJ/mol).

When the 1s wave functions of the two H

Atoms are linearly combined, we get a sigma (s) bonding orbital, denoted as s1s in the diagram here. This approach is called linear combination of atomic orbitals (LCAO), in the MO approach. In this approach, the sum of the two 1*s* orbitals (one for each atom) is the bonding orbital. In terms of wave mechanics, the two waves constructively interact. The difference of the two orbitals forms the antibonding orbital, б1s*, due to destructive interference. It is interesting to note that the anti bonding orbital is at a higher energy than the 1s atomic orbital. The energy level can be represented below:

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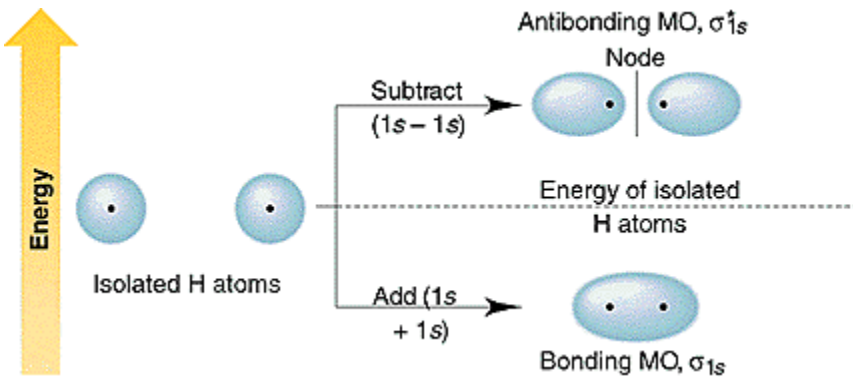

- Each orbital accommodates two electrons, and the two electrons in H–H fill the б1s molecularorbital (MO). Obviously, as a result of the formation of the H2 molecule, the energy of the system is lowered and becomes more stable.
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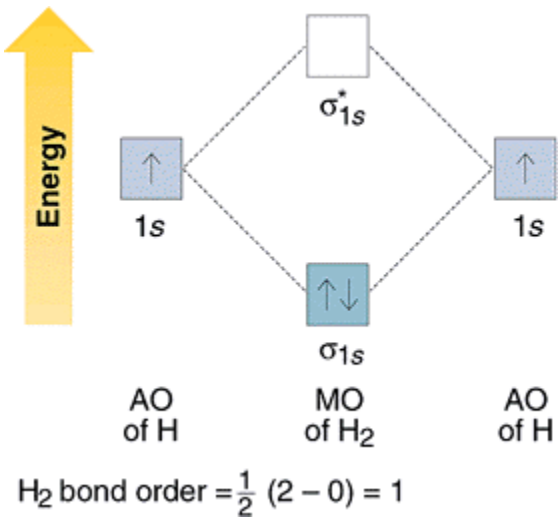

A molecular orbital energy-level diagram showing the relative energies of the atomic orbitals of atoms A and B (1*s* A and 1*s* B) and the bonding (1σ) and antibonding (2σ) molecular orbitals they form.

Molecular Orbitals of O2

The electronic configuration of oxygen (Z = 8) in the ground state is $1s^2, 2s^2\ 2p^4$. Each oxygen atomhas 8 electrons, hence, in O_2 molecule there are 16 electrons. Therefore, the electronic configuration of O_2 is as follows.

$$O_2 : KK(\sigma_{2s})^2(\sigma^*_{2s})^2(\sigma_{2px})^2(\pi_{2py},\pi_{2pz})^4(\pi^*_{2py},\pi^*_{2pz})^2$$

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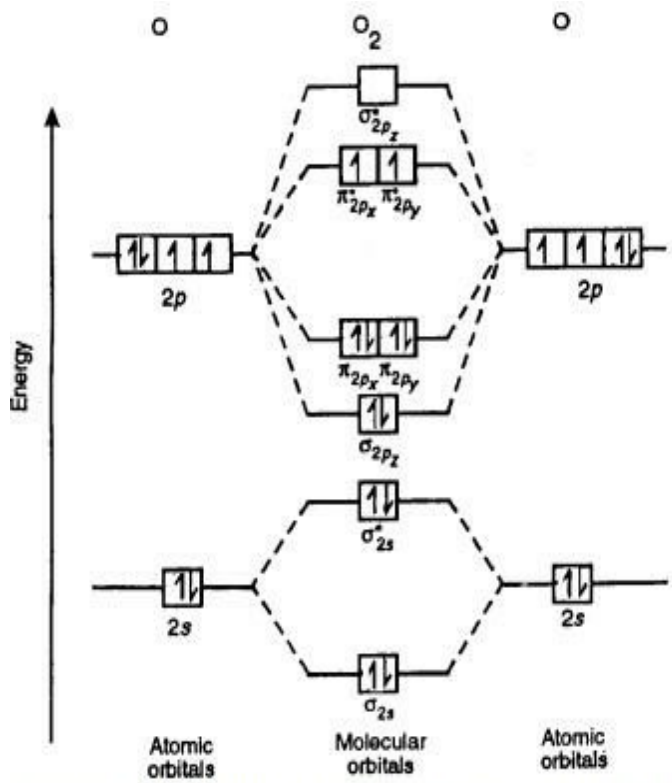
Here $(\sigma_{1s})^2 (\sigma^*1s)^2$ part of the configuration is abbreviated as KK. The molecular orbital energy level diagram of O_2 molecule is given in Fig.

Bond Order

The number of bonds between a pair of atoms is called the **bond order**. Bond orders canbe calculated from Lewis structures, which are the heart of the valence-bond model. Oxygen, for

example, has a bond order of two.
Bond order = $(N_b - N_a) / 2 = (8 - 4) / 2 = 2$.



Molecular orbital energy level diagram of O_2 molecule

Equations for atomic and molecular orbital.

Linear Combination of Atomic Orbitals (LCAO)

As per this method, the formation of orbitals is because of Linear Combination (addition or subtraction) of atomic orbitals which combine to form the molecule. Consider two atoms A and B which have atomic orbitals described by the wave functions Ψ_A and Ψ_B.
If the electron cloud of these two atoms overlaps, then we can obtain the wave function for the molecule by a linear combination of the atomic

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orbitals Ψ_A and Ψ_B. The below equation forms two molecular orbitals.

$$\Psi_{MO} = \Psi_A \pm \Psi_B$$

Bonding Molecular Orbitals

When the addition of wave function takes place, the type of molecular orbitals formed are Bonding Molecular Orbitals. We can represent them by $\Psi_{MO} = \Psi_A + \Psi_B$. They have lower energy than atomic orbitals involved.

Anti-Bonding Molecular Orbitals

When molecular orbital forms by the subtraction of wave function, the type of molecular orbitals formed are antibonding Molecular Orbitals. We can represent them as $\Psi_{MO} = \Psi_A - \Psi_B$. They have higher energy than atomic orbitals. Therefore, the combination of two atomic orbitals results in the formation of two molecular orbitals. They are the bonding molecular orbital (BMO) and the anti-bonding molecular orbital (ABMO).

Relative Energies of Molecular Orbitals

- **Bonding Molecular Orbitals (BMO)**

  Energy of Bonding Molecular Orbitals is less than that of Anti Bonding Molecular Orbitals. This is because of the increase in the attraction of both the nuclei for both the electron (of the combining atom).

- **Anti-Bonding Molecular Orbitals (ABMO)**

  Energy of Anti Bonding Molecular Orbitals is higher than Bonding Molecular Orbitals.
  This is because the electron tries to move away from the nuclei and are in a repulsive state.

Rules for Filling of Molecular Orbitals

There are certain rules while filling up molecular orbitals with electrons in order to write correct molecular configurations. They are

- **Aufbau Principle** – This principle states that those molecular orbitals which have the lowest energy are filled first.
- **Pauli's Exclusion Principle –** According to this principle, each molecular orbital can accommodate a maximum of two electrons having opposite spins.

Equations for atomic and molecular orbital

1. Electrons may be assigned to definite orbital : Wave-mechanical orbitals differ fundamentally from the precisely defined orbits of the Bohr quantum theory. The electron cannot be located exactly in the orbital

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(uncertainty principle), and one can only calculate the probability that the electron will be present in a given volume element in the region of the nucleus.

2. An electron in a particular orbital may be assigned definite energy.
3. Only two non-identical electrons may occupy a given orbital.
4. The force involve in chemical binding are electrostatic in nature.

Pi-molecular orbitals of benzene

According to this theory, all the C-atoms in benzene are sp^2-hybridized. Two sp^2-hybrid orbitals of each C-atom overlap with two sp^2-hybrid orbital of two other C-atoms to form sigma bonds. In this way there are six sigma bonds are formed between six C-atoms which are 120° apart. Remaining six sp^2-orbital of six C-atoms overlap with 1s orbital of six H-atoms individually to form six sigma bonds. Since sigma bond results from the overlap of above said planar orbital, all H and C atoms are in the same plane and their generate a hexagonal ring of C-atoms.

Each C-atom in benzene also has an unhybrid 2pz-orbital containing one electron. These 2pz-orbital are perpendicular to the plane of sigma bonds. Pz orbital in benzene

These 2pz-orbital by lateral overlapping form three alternate pi-bonds in benzene ring.

There are two possibilities of pi-bond formation in benzene.

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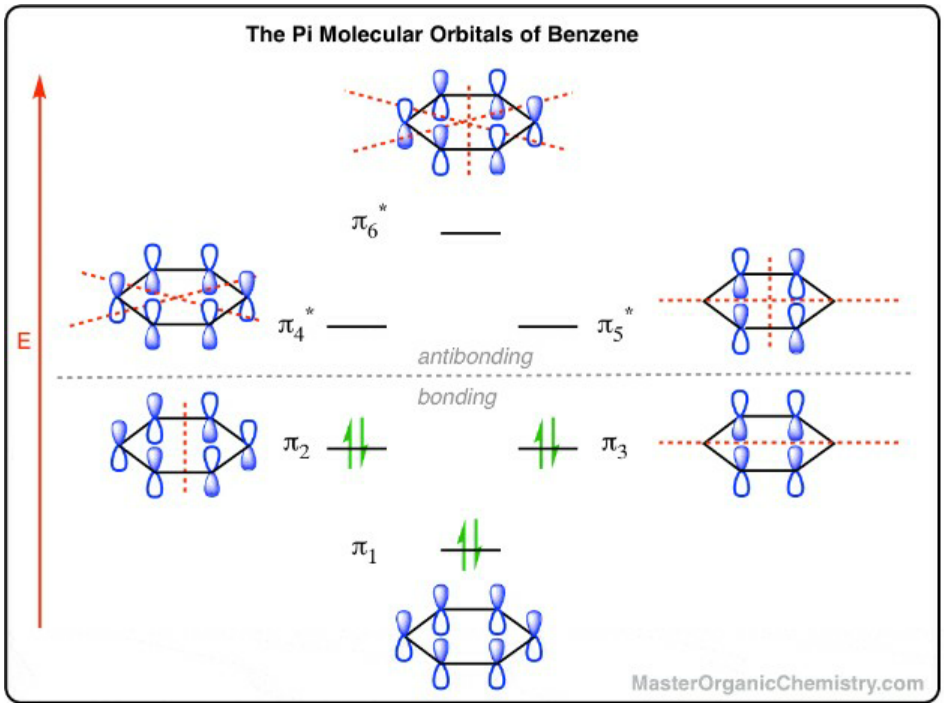

BOND LENGTH ANALYSIS IN BENZENE

C-C length in alkane = 1.54°A C=C length in alkene = 1.34°A But in benzene,

C-C length = 1.397°AC=C length = 1.397°A

This shows that in benzene single and double bonds have quite extraordinary character as they do not resemble to alkane and alkene in bond lengths. That is why benzene shows a behavior of saturated as well as an unsaturated hydrocarbon simultaneously. The orbital structure of benzene: All the carbon atoms in benzene are sp^2 hybridized. The three sp^2 hybrid orbitals are lying in one plane and oriented at an angle of 120°. The fourth unhybridized p- orbital having two lobes is lying perpendicular to the plane of the hybrid orbital. Two out of the three sp^2 hybrid orbitals of each carbon atom overlap axially with sp^2 hybrid orbitals of the neighboring carbon atoms on either side to form carbon-carbon σ bonds. The third hybrid orbital of each carbon atom overlaps axially with the half filled 1s- orbital of the hydrogen atom to form carbon-hydrogen sigma bonds. Thus there is six sigma C-C bonds and six sigma C-H bonds. There is still one unhybridized 2pz orbital on each carbon atom. This orbital consists of two lobes, one lying above and the other below the plane of the ring.

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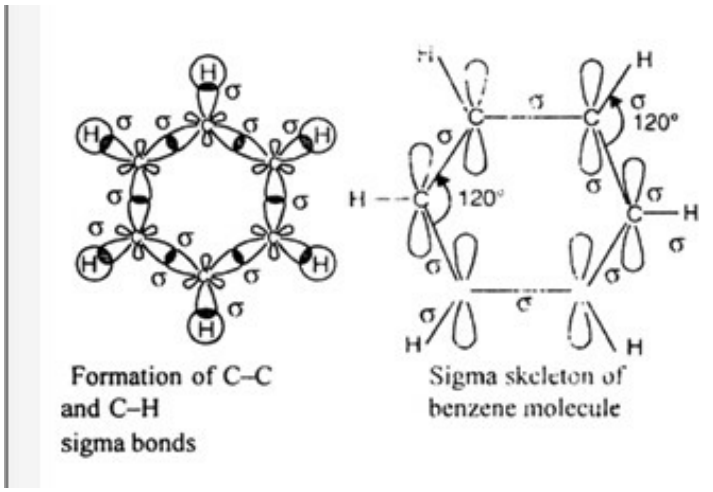
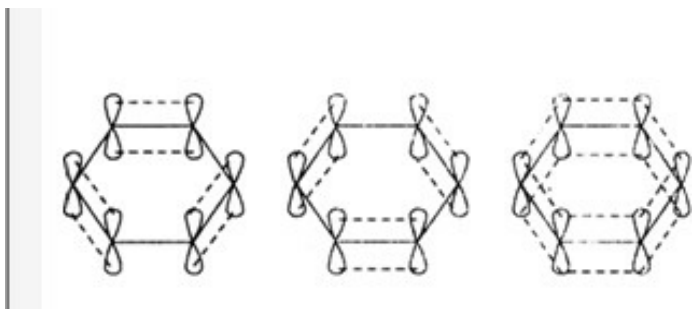


The unhybridized 2pz orbital on each carbon atom can overlap sidewise with the 2pzorbital of the two adjacent carbon atoms in two different ways as shown below giving riseto two sets of -bonds. Since 2pz orbital on any carbon atom can overlap sideways with the 2pz orbital on adjacent carbon atom on either side equally well, a continuous -molecular 3 orbitals will result which embraces all the six p-electrons as shown:



The net result is that there are two continuous rings-like electron clouds, one above and theother below the plane of atoms as shown. This delocalization of π-electrons imparts uniquestability to the benzene molecule.

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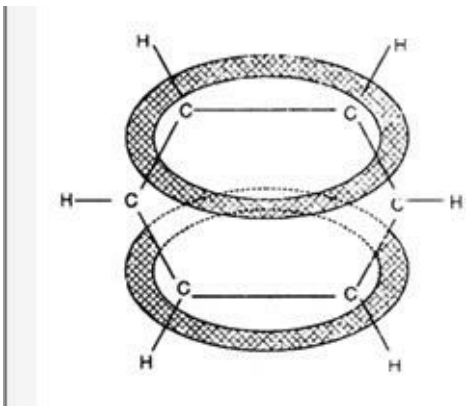
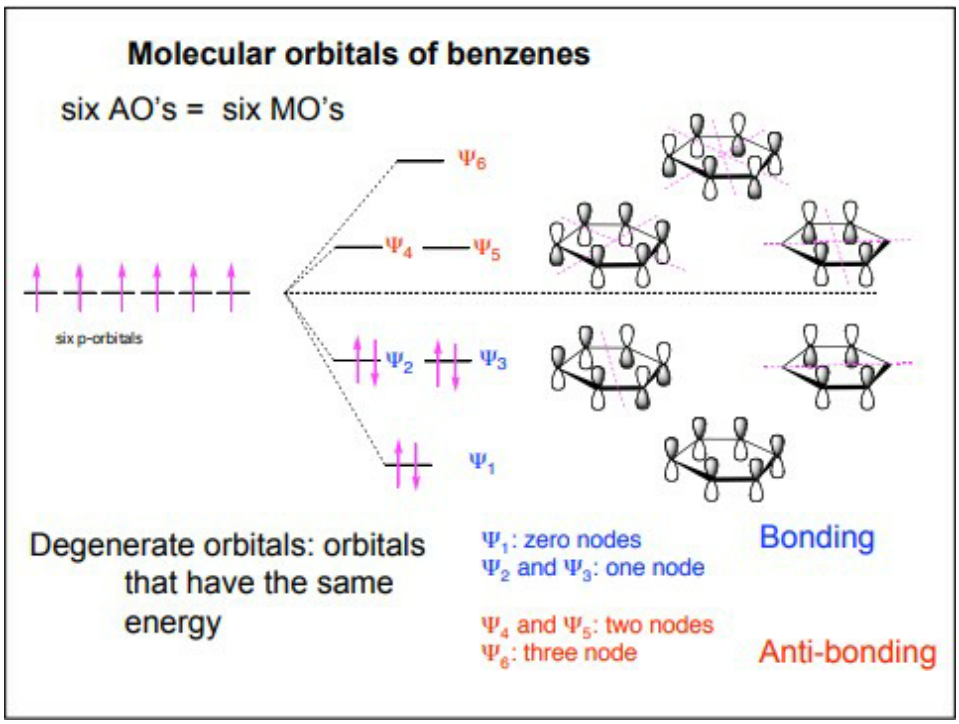


In benzene molecule all $\angle C-C-C$ and $\angle H-C-C$ are of 120° each and each C-C bond length is139 pm.



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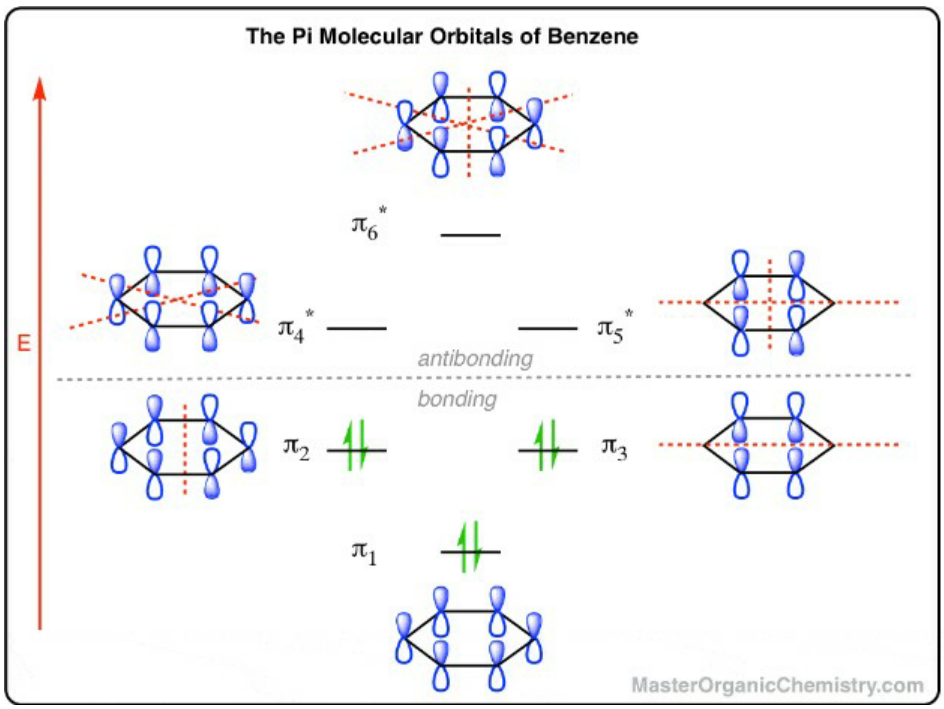

Question: Draw the molecular orbital diagram for the oxygen molecule, O_2. From this diagram, calculate the bond order for O_2. How does this diagram account for the paramagnetism of O_2?

Solution

Each oxygen atom contributes six electrons

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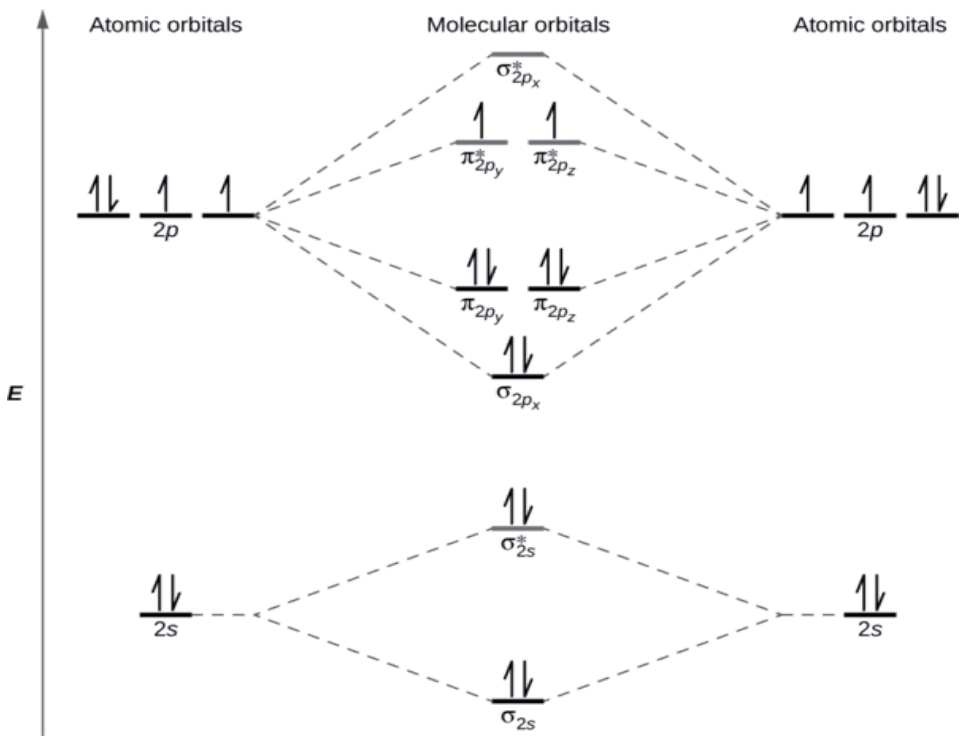

Figure 8.4.1 The molecular orbital energy diagram for O_2 predicts two unpaired electrons.

We calculate the bond order as;

$$O_2 = (8-4)/2 = 2$$

Oxygen's paramagnetism is explained by the presence of two unpaired electrons in the (π_{2py} , π_{2pz})* molecular orbitals.

Key Concepts and Summary

Molecular orbital (MO) theory describes the behaviour of electrons in a molecule in terms of combinations of the atomic wave functions. The resulting molecular orbitals may extend over all the atoms in the molecule. Bonding molecular orbitals are formed by in-phase combinations of atomic wave functions, and electrons in these orbitals stabilize a molecule. Antibonding molecular orbitals result from out-of-phase combinations of atomic wave functions and electrons in these orbitals make a molecule less stable. Molecular orbitals located along an internuclear axis are called σ MOs. They can be formed from *s* orbitals or from *p* orbitals oriented in an end-to-end fashion. Molecular orbitals formed from *p* orbitals oriented in a side-by-side fashion have electron density on opposite sides of the internuclear axis and are called π orbitals.

We can describe the electronic structure of diatomic molecules by applying molecular orbital theory to the valence electrons of the atoms. Electrons fill molecular orbitals following the same rules that apply to filling atomic orbitals; Hund's rule and the Aufbau principle tell us that lower-energy orbitals will fill first,

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electrons will spread out before they pair up, and each orbital can hold a maximum of two electrons with opposite spins. Materials with unpaired electrons are paramagnetic and attracted to a magnetic field, while those with all-paired electrons are diamagnetic and repelled by a magnetic field. Correctly predicting the magnetic properties of molecules is in advantage of molecular orbital theory over Lewis structures and valence bond theory.

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